

SAMARTH KALE

AI Systems Engineer · Computer Vision · Edge AI · Backend Systems

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Professional Summary

B.Tech Computer Engineering student (CGPA 8.80/10, MIT-ADT University, 2027) with internship experience spanning edge AI deployment and full-stack backend engineering. Shipped production AI systems end-to-end: YOLOv8 object detection fully on-device (Raspberry Pi 4, no cloud), a Django e-commerce platform with live payment processing, and an IoT-integrated home security pipeline. Published patent co-inventor; ranked 11th nationally at Smart India Hackathon 2025. Targeting AI Systems, Computer Vision, Edge AI, and backend-integrated AI roles.

Technical Skills

Languages: Python (primary), C, C++, SQL, JavaScript

AI & Computer Vision: YOLOv8, OpenCV, PyTorch, TensorFlow, Scikit-learn, face_recognition, Edge AI inference, Sensor Fusion

Backend & APIs: Django, Flask, FastAPI, REST API design, Razorpay payment gateway, Telegram Bot API, webhook handling

Cloud & DevOps: Git, GitHub, GitHub Actions, Docker, Linux (systemd, service config, boot automation), Render, ARM optimization

Embedded & Edge: Raspberry Pi 4/5, ESP32-CAM, I2C sensors, PIR, MQ135 gas sensor, on-device inference pipelines

Databases & Analytics: PostgreSQL, SQLite, schema design, NumPy, Pandas, Power BI

Internship Experience

AI Engineer Intern · CRiEYA · MIT-ADT University

May 2026 – Present

Python · YOLOv8 · OpenCV · Raspberry Pi 4 · I2C Sensors · Linux · ARM

- Developing real-time YOLOv8 object detection pipeline (100+ classes) on Raspberry Pi 4; optimized inference for low-latency edge processing with zero cloud dependency.
- Built OpenCV preprocessing pipeline handling frame capture, resolution scaling, and lighting normalization to sustain consistent inference under real-world conditions.
- Integrating multi-sensor hardware data (I2C) with vision output for sensor-assisted obstacle perception; configured Linux systemd services for auto-start deployment on boot.

Full Stack Development Intern · Rudra Agro Nursery

Jun 2025 – Jul 2025

Python · Django · REST APIs · PostgreSQL · Razorpay · Render · GitHub

- Led a 30-member engineering team as technical coordinator — owned GitHub branch strategy, PR reviews, and merge approvals across the full development cycle.
- Built and deployed a Django e-commerce platform in 30 days: REST API backend, PostgreSQL database, Razorpay payment gateway with webhook verification, deployed on Render with 99% uptime.
- Imported and cleaned a 1,000+ entry product catalog directly into the live database; platform processed 20+ successful test orders with 50+ daily active users during demo.

Projects

AI Smart Home Security & Automation System

Feb 2026 – May 2026

Python · ESP32-CAM · Raspberry Pi 5 · OpenCV · face_recognition · PIR · MQ135 · Telegram Bot API · Power BI

- Built face recognition system on ESP32-CAM using OpenCV and face_recognition library; achieved 92–95% identification accuracy across a 90-image dataset with support for 3 authorized users; triggers real-time Telegram Bot intruder alerts.
- Integrated PIR motion sensor and MQ135 gas sensor with automated buzzer and Telegram notification pipeline; average alert response time of 2–4 seconds end-to-end.
- Connected live event data to Power BI dashboard for security analytics; converts raw sensor readings into trend-tracked visual reports with full event logging.

AI-Based Blind Vision Assistive System

Sep 2025 – Jan 2026

Python · YOLOv8 · Flask · Raspberry Pi 4 · I2C Sensors · OpenCV · Linux

- Deployed on-device YOLOv8 inference pipeline on Raspberry Pi 4 — no cloud dependency; achieved ~15 FPS throughput with ~90% obstacle detection accuracy across 80 object classes.
- Integrated I2C sensor fusion with vision output and Flask monitoring backend; validated end-to-end pipeline across 50+ real-world navigation scenarios with 5+ peer users.
- Engineered voice alert pipeline with ~0.7s average latency; optimized guidance logic for 15 high-priority obstacle categories across 80 supported classes.
- Wrote Python scripts for data streaming, structured logging, and error handling; deployed as a persistent Linux service with systemd auto-restart.

Education

B.Tech, Computer Engineering — MIT-ADT University, Pune

Expected May 2027

CGPA: 8.80 / 10 · No active backlogs · Specialization: AI & Edge Computing

Achievements

- **Published Patent** — Co-inventor, "IoT and Blockchain Based Smart Basket System for Supply Chain Tracking" (No. 202521133452A). Built and debugged the complete hardware-software prototype including sensor integration, Python firmware, and end-to-end implementation pipeline.
- **Smart India Hackathon 2025** — Ranked 11th nationally in Agriculture & Rural Development domain (44th among all software teams). Delivered a working AI-powered agriculture booth — soil health analysis, livestock monitoring, digital land records, and government scheme access — during a 36-hour national competition.